Employee Management System Database Schema and Database Code Submission

1. Introduction and Application Relevance

The chosen application area is an Employee Management System (EMS) designed for HR teams and managers. It is relevant because organizations require a structured and reliable system to manage employee profiles, department assignments, attendance tracking, leave management, and payroll processing.

A relational database system is ideal for this application because it ensures:

Data integrity

Strong relationships between entities

Accurate reporting capabilities

Reduced redundancy through normalization

This database design supports:

Employee record management

Department and manager assignment

Daily attendance tracking

Leave request lifecycle (Pending, Approved, Rejected)

Monthly payroll generation and analytics

2. Database Schema Design (Normalized)

The schema is designed using Third Normal Form (3NF) to reduce redundancy and improve maintainability.

Tables Included

app_users

departments

employees

attendance_logs

leave_requests

payroll_records

Each table defines primary keys, foreign keys, and constraints to ensure strong data integrity.

3. Table Descriptions 3.1 app_users

Stores login and account-level data along with user roles.

Primary Key

user_id

Constraints

Unique email address

Valid role values (Admin, HR, Manager, Employee)

Purpose

Separates authentication data from employee profile information.

3.2 departments

Stores department-level information within the organization.

Primary Key

department_id

Foreign Key

manager_user_id   app_users.user_id

Constraints

Unique department name

Purpose

Maintains department data and identifies department managers.

3.3 employees

Stores employee HR profile information.

Primary Key

employee_id

Foreign Keys

user_id   app_users.user_id

department_id   departments.department_id

Constraints

Unique user_id

Unique phone number

Salary must be greater than zero

Valid employment status values

Purpose

Links employee identity with system user accounts and departments.

3.4 attendance_logs

Stores daily attendance records.

Primary Key

attendance_id

Foreign Key

employee_id   employees.employee_id

Constraints

One record per employee per day

Valid attendance status

check_out ≥ check_in

Purpose

Tracks employee attendance and working hours.

3.5 leave_requests

Stores leave applications submitted by employees.

Primary Key

leave_id

Foreign Key

employee_id employees.employee_id

Constraints

Valid leave type

Valid leave status

end_date ≥ start_date

Derived Column

days_requested

Purpose

Manages leave requests and approval workflow.

3.6 payroll_records

Stores employee payroll details.

Primary Key

payroll_id

Foreign Key

employee_id employees.employee_id

Constraints

One payroll record per employee per month

Salary components must be non-negative

Derived Column

net_salary

Purpose

Stores salary breakdown and monthly payment information.

4. Normalization Notes

The database follows Third Normal Form (3NF) principles.

Key normalization decisions:

User login data is stored separately from employee profile data.

Department information is stored once and referenced through foreign keys.

Attendance, leave, and payroll are stored as transactional tables referencing employees.

Repeating groups are avoided.

Relationships are maintained using foreign keys.

This design ensures data consistency, reduced redundancy, and easier maintenance.

5. Entity Relationship (ER) Diagram

The ER diagram source is provided using Mermaid syntax.

File Included

er-diagram.mmd

The diagram illustrates relationships between:

Users

Departments

Employees

Attendance Logs

Leave Requests

Payroll Records

Rendering Instructions

To render the ER diagram:

Open Mermaid Live Editor

Upload the .mmd file

Export the diagram as PNG or SVG

6. Database Implementation

The database is implemented using PostgreSQL.

SQL File Included

employee_management_db.sql

This SQL file contains:

Database Setup

DROP TABLE statements for clean reruns.

Table Creation

CREATE TABLE statements

Primary keys

Foreign keys

Unique constraints

Check constraints

Sample Data

Each table contains at least five sample records.

7. Data Manipulation and Querying

The SQL script includes all required query types.

A. Data Manipulation Queries

The following operations are demonstrated:

DML 1 — Insert

Insert a new user and linked employee record using a Common Table Expression (CTE).

DML 2 — Update

Update employee salary details.

DML 3 — Delete

Delete an attendance record.

B. Data Retrieval Queries

At least five retrieval queries are included.

Q1 — Employee Directory

Displays employees with department and role information.

Features used:

JOIN

ORDER BY

Q2 — Department Headcount

Displays number of employees in each department.

Features used:

GROUP BY

COUNT

Q3 — Monthly Payroll Totals

Calculates total payroll expenses.

Features used:

GROUP BY

Aggregate functions

Q4 — Attendance Summary

Shows attendance statistics per employee.

Features used:

JOIN

Conditional aggregation

Q5 — Leave Requests with Department Details

Displays leave requests with department information.

Features used:

JOIN

FILTER

ORDER BY

C. Complex Queries

Two advanced SQL queries are included.

CQ1 — Highest Paid Employee per Department

Uses:

Common Table Expression (CTE)

Window Function (ROW_NUMBER)

Purpose:

Identify the highest-paid employee in each department.

CQ2 — Employees Earning Above Department Average

Uses:

Correlated subquery

Purpose:

Identify employees whose salary is above the department average.

8. How to Run the Database

Follow these steps to run the project.

Step 1 — Create Database

Create a PostgreSQL database.

Example:

CREATE DATABASE ems_db;

Step 2 — Open SQL Client

Use one of the following tools:

pgAdmin

DBeaver

psql

Step 3 — Execute SQL Script

Run the SQL file:

employee_management_db.sql

Step 4 — Execute Queries

Run each query section to view results.

9. Evidence / Screenshots to Include

Add screenshots in the final report showing:

Successful execution of CREATE TABLE statements

Data preview for each table (5+ rows)

Output of INSERT, UPDATE, DELETE operations

Output of all retrieval queries (Q1–Q5)

Output of complex queries (CQ1–CQ2)

Rendered ER diagram

Recommended Screenshot Names

Screenshot_01_CreateTables.png

Screenshot_02_UsersData.png

Screenshot_03_EmployeesData.png

Screenshot_04_RetrievalQueries.png

Screenshot_05_ComplexQueries.png

Screenshot_06_ERDiagram.png

10. Mapping to Evaluation Criteria Correctness and Efficiency

Normalized schema design

Strong key relationships

Strict integrity constraints

Implementation Quality

Executable PostgreSQL script

Clear separation of DDL, DML, and query sections

Complexity and Creativity

Window functions

Correlated subqueries

Documentation and Presentation

Structured documentation

Evidence checklist

ER diagram visualization

Best Practices

Consistent naming conventions

Appropriate data types

Clean SQL formatting

11. Files Included in Submission Folder

The final submission ZIP folder includes:

employee_management_db.sql

er-diagram.mmd

Database_Documentation.md